RealWaste Classification: Machine Learning for Waste Sorting

Project category: Applied Machine Learning | Computer Vision

Team members

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Motivation

Manual sorting is labor-intensive and inconsistent when waste is dirty, damaged, or mixed. Image classification could support recycling, organic-waste diversion, and landfill audits. The RealWaste paper shows the value of authentic images: its best model trained on pristine materials reached only 49.69% accuracy on landfill test images, versus 89.19% when trained on RealWaste [1]. We will study the nine categories in the supplied 4,752-image project subset.

Method

We will train a small CNN on RealWaste from scratch with 3-4 convolutional layers, batch normalization, dropout, augmentation, and the Adam optimizer to demonstrate backpropagation and regularization (CO3). Classical baselines will use HOG and color-histogram features with logistic regression, linear and RBF-kernel SVMs, and random forest. We will then retrain the repository's MobileNetV3, EfficientNetB0, and GoogLeNet with our own split and fixed seeds to compare pretrained models against the scratch CNN. YOLOv8n classification provides an additional comparison. A detection experiment requires separate, reliable bounding-box annotations.

Intended experiments and expected results

Every classifier will use the same stratified train/validation/test split after duplicate checks. We will report top-1 accuracy, macro F1, per-class precision, recall and F1, and confusion matrices; macro F1 is important because plastic has 921 images while textile trash has 318. The majority-class baseline is about 19.4% accuracy. We will compare classical, scratch, and pretrained models, test augmentation and layer-freezing choices, and inspect example misclassified images, especially miscellaneous trash, plastic, and paper, with explanations of likely errors. We expect pretrained models to outperform the scratch and classical baselines and approach the high-80% accuracy range reported in the supplied project summary. Any detection results will use detection metrics separately.

Proposed member contributions

Harin Savaliya will coordinate experiments and integrate the report. Rushikesh Talaviya will audit labels and prepare the shared split and features. Divyang Vanjara will train the scratch CNN and classical baselines. Akshat Agrawal will retrain the pretrained CNNs and YOLO classifier. Anish Reddy will calculate metrics, investigate misclassified images, and prepare the comparison and demo. These roles are proposed responsibilities.

[1] S. Single, S. Iranmanesh, and R. Raad, "RealWaste: A Novel Real-Life Data Set for Landfill Waste Classification Using Deep Learning," Information 14(12), 633, 2023. doi:10.3390/info14120633.